INCH-POUND

MIL-DTL-4088L
1 April 2021
SUPERSEDING
MIL-W-4088K
21 November 1988

DETAIL SPECIFICATION

WEBBING, TEXTILE, WOVEN NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers woven nylon, textile webbing.

1.2. Classification. The nylon webbing will be available in the types and classes as specified (see 6.2). The types will conform to the requirements as specified in Table IV and V as applicable for the class specified.

1.2.1 Classes.

Class 1	-	Shuttle loom, nylon 6,6
Class 1A	-	Shuttleless loom, nylon 6,6
Class 2	-	Shuttle or shuttleless loom, nylon 6 or nylon 6,6

1.2.1.1 Class reference. When procurement documents referencing this specification do not specify the class of webbing, any class in paragraph 1.2 may be used.

Comments, suggestions, or questions on this document should be addressed: ATTN: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil

AMSC N/A FSC 8305

2. APPLICABLE DOCUMENTS

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2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1480 - Color Codes for Webbing, Textile; Manufacturers Identification
MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COLOR ASSOCIATION OF THE UNITED STATES (CAUS)

(Copies of this document are available at https://www.colorassociation.com.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

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AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY

ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1423/D1423M - Standard Test Method for Twist in Yarns by Direct-Counting
ASTM D1776/D1776 - Standard Practice for Conditioning and Testing Textiles
ASTM D1777 - Standard Test Method for Thickness of Textile Materials
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3774 - Standard Test Method for Width of Textile Fabric
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling)Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D6770 - Standard Test Method for Abrasion Resistance of Textile Webbing (Hex Bar Method)
ASTM D6775 - Standard Test Method for Breaking Strength and Elongation of Textile Webbing, Tape and Braided Material

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1. First article. When specified (see 6.2) a sample of the webbing shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished dyed webbing shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Nylon yarn. The webbing yarns shall be made from bright, high tenacity, light and heat resistant nylon. The Class 1 and Class 1A webbing shall use nylon 6,6 and Class 2 webbing shall use either nylon 6 or nylon 6,6. The yarn shall not be bleached. Testing shall be as specified in 4.5.

3.4.1.1 Denier and filament. The yarn used in the manufacturer of the webbing shall be of the denier and filament specified in Tables IV or V, before dyeing, except for the identification yarns (see 3.7). The following yarn substitution may be made for all webbings listed in Table IV and Table V: 840/3 = 2520/1, 840/3 = 1260/2, 840/2 = 1680/1, 420/3 = 1260/1.

3.4.1.2 Twist. The warp, binder and filling yarns shall have a minimum of 2-1/2 turns per inch in the final twist whether single or plied, except types XXII and XXVII which shall have a minimum of 1-1/2 turns per inch. The number of single yarns specified in Tables IV and V shall be twisted together in one operation.

3.5 Color. Unless otherwise specified (see 6.2), the webbing shall be furnished in the following colors:

a. Natural White
b. Natural White (special purpose items)
c. Olive Drab 7
d. Camouflage Green 483
e. Tan 499
f. Air Force Sage Green 34158 (AMS-STD-595)
g. Air Force Yellow 13655 (AMS-STD-595)

3.5.1 Dyeing. When dyed webbing is specified, the webbing shall be yarn or piece dyed.

3.5.2 Visual shade matching. The color and appearance of the webbing shall visually match the applicable color standard or the standard sample when tested as specified in 4.5.

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3.5.2 Colorfastness. The finished dyed webbing and the identification yarns (see 3.7) shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements for dyed webbing and identification yarns.

	Colorfastness to laundering (3 cycles) (Color c Change and Staining)	Colorfastness to crocking (Wet and Dry)	Colorfastness to light (20 AFU or 85 kJ/(m^2nm) @ 420 nm)
Webbing nylon body yarns	2-3	3.5	3-4
Identification yarns	2-3	-	3-4

1/ AFU - AATCC Fading Units

3.6 Spectral reflectance.

3.6.1 Spectral reflectance, Tan 499 and Camouflage Green 483. The reflectance values shall conform to the requirements listed below, in Table II when tested as specified in 4.5.

TABLE II. Spectral reflectance requirements, Tan 499 and Camouflage Green 483.

	Reflectance			
Wavelength, Nanometers (nm)	Tan 499		Camouflage Green 483	
	Min.	Max.	Min.	Max.
600	8	26	3	10
620	8	26	3	10
640	8	30	3	10
660	8	34	3	11
680	12	38	3	13
700	12	40	4	28
720	16	46	5	40
740	22	50	7	52
760	30	50	11	60
780	34	54	17	64
800	36	56	24	67
820	38	58	32	70
840	38	58	37	71
860	40	60	40	73

3.7 Identification yarns. Colored yarns shall be woven into the webbing, as specified herein, to permanently identify the webbing by type and manufacture, and when shuttleless construction and/or nylon 6 is utilized, except that natural white (special purpose items) webbing shall have no

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colored yarn ends (see 6.2). When webbing is piece dyed, polyester yarn of approximate denier to the nylon body of the webbing shall be used for all required colored yarns except when black is specified, the identification yarn shall be 200 or 210 or 400/2 denier nylon 6 or 6, 6, color-sealed black or 70, 90, or 150 denier polyester, color-sealed black (see Table III). The colorfastness for the identification yarns shall be as specified in Table II.

3.7.1 Type identification. The identification of type shall be as specified in Table III. The color of the identification yarns shall match the cable numbers of the Color Association of the United States, Inc.

3.7.2 Other types. When type identification is required for other types, the identification yarns shall be as specified in the procurement documents (see 6.2).

TABLE III. Identification of yarn type.

Type	Color	Cable No.	Yarns and location
VI	Red	80081	2 at center of warp
VII 1/	Yellow	80068 2/	2 at each selvage
VIII	Black	85018	2 at center of warp
XII	Red	80081	1 at each selvage
XIII 1/	Black	85018	2 at each selvage
XIX	Green	80063	2 at center of warp
XXII	Black	85018	2 at each selvage and 2 at the center
XXVI	Yellow	80068 2/	2 at center of warp
XXVII	Black	85018	1 at each selvage

1/ These being double plain weaves, the identification yarns shall weave so that 1 end shall show on the face and 1 on the back at each selvage
2/ Any yellow shade between Cable no. 80068 and Color chip no. 23793 of AMS-STD-595 is acceptable.

3.7.3 Manufacturer's identification. Each manufacturer of Types VII, IX, X and XIII shall incorporate, as part of the binder warp, 2 ends dyed to match the shade assigned to the manufacturer in accordance with MIL-STD-1480 (see 6.2 and 6.4).

3.7.4 Shuttleless loom identification. When shuttleless loom construction is utilized for Classes 1A and 2, the catch-cord shall be 70 or 210 or 400 denier nylon, color-sealed black. Alternatively, the catch-cord shall be 70, 90 or 150 denier polyester color-sealed.

3.7.5 Nylon 6 identification. When nylon 6 is utilized for Class 2, one (1) edge of the webbing shall contain a red marker yarn, Cable no. 80081.

3.8 Weave. The weaves shall be as specified in 3.8.1 through 3.8.8. The filling yarn of Classes 1A and Class 2 webbing shall traverse the full width of the webbing and shall be held at the edge by an extra catch-cord end interlaced with the filling yarn in accordance with the method shown in Figures 7 or 8.

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3.8.1 Weave for Types I, Ia, Ib, II, III, IV, VI, VIII, VIIIa, VIIIb, VIIIc, XII, XV, XVI and XVII. The weaves for Types I, Ia, Ib, II, III, IV, VI, VIII, VIIIa, VIIIb, VIIIc, XII, XV, XVI and XVII shall be 2 up, 2 down herringbone twill and 1 reversal at the center of the webbing.

3.8.2 Weave for Types VII, IX, X, XIII, XXV, and XXVIII. The weave for Types VII, IX, X, XIII, XXV, and XXVIII shall be a double plain weave. Separate binder warp ends shall weave 2 up, 2 down 1 end as 1. All other warp yarns shall weave 2 ends as 1 except that the edge warp yarns shall weave 1 end as 1 not exceeding 8 ends on 1 selvage and 9 on the other.

3.8.2.1 Alternate for the Types VII, IX, X and XIII. An acceptable alternate weave for the Types VII, IX, X and XIII webbings shall be woven as stated above except the Manufacturers' Identification Yarns shall weave as a stuffer yarn between two adjacent binder yarns.

3.8.3 Weave for Types XIV, XVIII, XIX and XXI. The weave for Types XIV, XVIII, XIX and XXI shall be a 5 up, 1 down, 1 up, 5 down herringbone twill with 1 reversal at the center of the webbing. The weave is shown on Figure 1. Both the body and the selvage warp shall weave 1 end as 1.

3.8.4 Weave for Types XX and XXIII. The weave for Types XX and XXIII shall be as shown on Figure 2.

3.8.5 Weave for XXII. The weave for Type XXII shall be as shown on Figure 3.

3.8.6 Weave for Type XXIV. The weave for Type XXIV shall be 2 up and 2 down herringbone twill with 3 reversals as shown on Figure 4.

3.8.7 Weave for Type XXVI. The weave for Type XXVI shall be a 5 up, 1 down, 5 down, 1 up, 1 down (7 harness + 4 selvage = 11) twill with no reversal of the weave at the center. The weave shall be as shown on Figure 5.

3.8.8 Weave for Type XXVII. The weave for Type XXVII shall be as shown on Figure 6.

3.9 Physical requirements. The construction and physical requirements of the finished webbing shall be as specified in Tables IV and V when tested as specified in 4.5.

3.9.1 Breaking strength. The original breaking strength of any specimen shall be no lower than the minimum specified in Tables IV and V.

3.9.1.1 Breaking strength after abrasion (Types XXII and XXVII). The Types XXII and XXVII webbing shall retain no less than 90 percent of the applicable minimum breaking strength values specified in Table IV when tested as specified in 4.5.

3.9.2 Curvature. The finished webbing shall show no more lateral curvature than 1/4 inch within a yard when tested as specified in 4.5.

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TABLE IV. Physical requirements, Class 1.

Type 1/	Width (inches)	Thickness (inches)	Weight (ounces per linear yard) (max.)	Breaking strength, lbs. (min.)	Ends in warp (min.)		Picks per inch (min.)	No. of single yarns for final plied yarns (min.)			Yarn denier and filament before dyeing
					Face and back	Binder		Warp	Binder	Filling	
I	9/16 (±1/32)	0.025-0.040	0.28	500	92		34	1		1	420/68W 840/140F
Ia	3/4 (±1/32)	0.025-0.035	0.32	600	108		34	1		1	420/68W 840/140F
Ib	3/8 (±1/32)	0.025-0.040	0.28	340	58		34	1		1	420/68W 840/140F
II	1 (±1/32)	0.025-0.040	0.42	600	134		34	1		1	420/68W 840/140F
III	1-1/4 (±1/32)	0.025-0.040	0.52	800	168		34	1		1	420/68W 840/140F
IV	3 (±1/8)	0.025-0.040	1.20	1,800	400		34	1		1	420/68W 840/140F
VI	1-23/32 (±1/16)	0.030–0.050	1.15	2,500	114		21	2		2	840/140 W F
VII	1-23/32 (±1/16)	0.060-0.100	2.35	6,000	229	27	26	2	1	2	840/140 W B F
VIII	1-23/32 (±1/16)	0.040-0.070	1.60	4,000	166		18	2		2	840/140 W F
VIIIa	3 (±3/32)	0.040-0.070	2.80	6,300	280		18	2		2	840/140 W F
VIIIb	2 (±1/16)	0.040-0.070	1.80	4,500	192		18	2		2	840/140 W F
VIIIc	2-1/4 (±1/16)	0.040-0.070	2.10	5,300	222		18	2		2	840/140 W F
IX	3 (±3/32)	0.065-0.100	4.00	9,000	257	31	28	3	2	2	840/140 W B F
X	1-23/32 (±3/32)	0.105-0.140	3.70	9,500	257	31	22	3	1	2	840/140 W B F
XII	1-23/32 (±1/16)	0.025-.040	0.85	1,200	266		34	1		1	420/68W 840/140F

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TABLE IV. Physical requirements, Class 1. - Continued

Type 1/	Width (inches)	Thickness (inches)	Weight (ounces per linear yard) (max.)	Breaking strength, lbs. (min.)	Ends in warp (min.)		Picks per inch (min.)	No. of single yarns for final plied yarns (min.)			Yarn denier and filament before dyeing
					Face and back	Binder		Warp	Binder	Filling	
XIII	1-23/32 (±3/32)	0.080-0.120	2.90	7,000	281	34	24	2	1	2	840/140 W B F
XIV	½ (±1/32)	0.070-0.100	0.80	1,200	91		36	7		7	210/34 W F
XV	2 (±1/16)	0.035-0.050	1.25	1,500	88		15	2		2	840/140 W F
XVI	1-23/32 (±1/16)	0.045-0.080	2.00	4,500	198		17	2		2	840/140 W F
XVII	1 (±1/16)	0.045-0.070	1.15	2,500	114		15	2		2	840/140 W F
XVIII	1 (±1/16)	0.100-0.160	2.05	6,000	260		18	2		2	840/140 W F
XIX	1-3/4 (±3/32)	0.100-0.130	4.10	10,000	280		18	3		2	840/140 W F
XX	1 (±3/32)	0.170-0.210	3.25	9,000	162	26	19	5	1	3	840/140W B F
XXI	1-1/4 (±1/16)	0.065-0.085	1.70	3,600	260		25	5		10	210/34 W F
XXII	1-23/32 (±3/32)	0.090-0.120	3.50	9,500	259		18	3		2	840/140 W F
XXIII	1-1/8 (±3/32)	0.200–0.300	3.70	12,000	324	27	15	3	2	3	840/140 W B F
XXIV	1-15/16 (±3/32)	0.055-0.075	2.25	5,500	244		17	2		3	840/140 W F
XXV	1 (±1/16)	0.080-0.125	1.50	4,500	169	20	22	2	1	2	840/140 W B F
XXVI	1-3/4 (±1/16)	0.150-0.180	4.90	15,000	236		16	5		3	840/140 W F
XXVII	1-23/32 (±1/16)	0.085-0.110	2.90	6,500	215		24	3		2	840/140 W F
XXVIII	2-1/4 (±3/32)	0.080-0.110	3.80	8,700	257	31	22	3	1	2	840/140 W B F

1/ Types V and XI have been deleted.

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TABLE V. Physical requirements, Class 1A and 2.

Type 1/	Width (inches)	Thickness (inches)	Weight (ounces per linear yard) (max.)	Breaking strength, lbs. (min.)	Ends in warp (min.)		Picks per inch (min.) 2/	No. of single yarns for final plied yarns (min.)			Yarn denier 3/ Warp, binder and filling
					Face and back	Binder		Warp	Binder	Filling	
I	9/16 (±1/32)	0.025-0.040	0.28	500	92		68	1		1	420
Ia	3/4 (±1/32)	0.025-0.035	0.32	600	108		68	1		1	420
II	1 (±1/32)	0.025-0.040	0.42	600	134		68	1		1	420
III	1-1/4 (±1/32)	0.025-0.040	0.52	800	168		68	1		1	420
IV	3 (±1/8)	0.025-0.040	1.20	1,800	400		68	1		1	420
VI	1-23/32 (±1/16)	0.030–0.050	1.15	2,500	114		42	2		1	840
VII	1-23/32 (±1/16)	0.060-0.100	2.35	6,000	229	27	52	2	1	1	840
VIII	1-23/32 (±1/16)	0.040-0.070	1.60	4,000	166		36	2		1	840
VIIIa	3 (±3/32)	0.040-0.070	2.80	6,300	280		36	2		1	840
VIIIb	2 (±1/16)	0.040-0.070	1.80	4,500	192		36	2		1	840
VIIIc	2-1/4 (±1/16)	0.040-0.070	2.10	5,300	222		36	2		1	840
IX	3 (±3/32)	0.065-0.100	4.00	9,000	257	31	56	3	2	1	840
X	1-23/32 (±3/32)	0.105-0.140	3.70	9,500	257	31	44	3	1	1	840
XII	1-23/32 (±1/16)	0.025-.040	0.85	1,200	266		68	1		1	420
XIII	1-23/32 (±3/32)	0.080-0.120	2.90	7,000	281	34	48	2	1	1	840
XIV	1/2 (±1/32)	0.070-0.100	0.80	1,200	91		70	7		4	210
XV	2 (±1/16)	0.035-0.050	1.25	1,500	88		30	2		1	840

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TABLE V. Physical requirements, Class 1A and 2. - Continued

Type 1/			Weight (ounces per linear yard) (max.)	Breaking strength, lbs. (min.)	Ends in warp (min.)		Picks per inch (min.) 2/	No. of single yarns for final plied yarns (min.)			Yarn denier 3/ Warp, binder and filling
					Face and back	Binder		Warp	Binder	Filling	
XVI	1-23/32 (±1/16)	0.045-0.080	2.00	4,500	198		34	2		1	840
XVII	1 (±1/16)	0.045-0.070	1.15	2,500	114		30	2		1	840
XVIII	1 (±1/16)	0.100-0.160	2.05	6,000	260		36	2		1	840
XIX	1-3/4 (±3/32)	0.100-0.130	4.10	10,000	280		36	3		1	840
XX	1 (±3/32)	0.170-0.210	3.25	9,000	162	26	38	5	1	3	840/420 4/
XXI	1-1/4 (±1/16)	0.065-0.085	1.70	3,600	260		50	5		5	210
XXII	1-23/32 (±3/32)	0.090-0.120	3.50	9,500	259		36	3		1	840
XXIII	1-1/8 (±3/32)	0.200-0.300	3.70	12,000	324	27	30	3	2	3	840/420 4/
XXIV	1-15/16 (±3/32)	0.055-0.075	2.25	5,500	244		34	2		3	840/420 4/
XXV	1 (±1/16)	0.080-0.125	1.50	4,500	169	20	44	2	1	1	840
XXVI	1-3/4 (±1/16)	0.150-0.180	4.90	15,000	236		32	5		3	840/4204/
XXVII	1-23/32 (±1/16)	0.085-0.110	2.90	6,500	215		48	3		1	840
XXVIII	2-1/4 (±3/32)	0.080-0.110	3.80	8,700	257	31	44	3	1	1	840

1/ Types V and XI have been deleted.
2/ Two (2) picks per shed.
3/ The use of Nylon 6 or 6,6 is allowed for Class 2. A combination of these fibers in a single webbing shall not be allowed.
4/ 840 denier warp and binder, 420 denier filling.

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3.10 Residual shrinkage. The residual shrinkage of natural and dyed Type Ib webbing shall be not greater than 2.0 percent when tested as specified in 4.5.

3.11 pH. The pH of the water extract of the finished webbing shall be no less than 5.0 and no greater that 8.5 when tested as specified in 4.5

3.12 Length and put-up. The webbing shall be put-up in rolls. Unless otherwise specified (see 6.2), the length of the rolls shall be as follows:

Types XVIII and XXVI	- Not less than 90 yards nor more than 110 yards with not more than one (1) piece per roll.
Types XX and XXIII	- Not less than 60 yards nor more than 70 yards per roll. A roll shall contain no more than two (2) pieces and each piece shall be not less than 10 yards in length.
Type Ib	Not less than 490 yards or more than 500 yards per roll. A roll shall contain no more than 15 pieces and each piece shall be not less than 10 yards in length.
All other types	- Not less than 90 yards nor more than 110 yards. A roll shall contain no more than three (3) pieces and each piece shall not be less than 10 yards in length.

3.13 Identification tickets. Each roll shall have a ticket attached with not finer than 5-ply cotton string doubled to not less than 6 inches long. The tickets shall be made of paperboard not less than 0.015-inch in thickness and the color shall be manila or light in intensity to permit easy reading of printed, stamped or typed markings. The use of handwritten entries is prohibited. The ticket shall have clipped corners at the end where a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying string. The ticket shall be legibly printed with water insoluble ink with the following information:

Stock number
Nomenclature
Specification number
Yardage
Contract number and date of manufacture (month and year)
Contractor's name
Name of contracting agency.
When webbing is not procured directly by the Government, the above information shall appear on the container, in which case the roll tickets are not required

3.14 Fiber identification. Each webbing roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.15 Workmanship. The finished webbing shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of the referenced documents.

4.4 Visual examination. Each roll in the sample shall be examined yard by yard for defects in accordance with MIL-STD-3064, Type IV, except the defects specified in Table VI shall also be scored four (4) points for each yard in which they occur.

TABLE VI. Defects that score four (4) points.

Defect Description
Two (2) yarns per shed (Class 1 only)
Drop-ply - clearly visible on more than 2 ends within the same length extending over nine (9) linear inches or more
Jerked-in filling, slough-off, slug, slub - clearly visible loop or filling pulled in at edges
Mispick, double pick - two or more across the full width; single across the full width
Slack end - two or more ends jerked in between picks, forming clearly visible loops on the surface. Single jerked in between picks or forming clearly visible loops on the surface
Any smash
Dropped knitted stitch on edge; catch-cord missing
Tight end visible up to 12 inches
Wrong draw extending for more than 9 inches

4.4.1 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for defects in roll identification and marking as specified in MIL-STD-3064.

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4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 Component and end item testing. The yarn and webbing shall be tested for the characteristics listed in Table VII. The methods of testing specified, wherever applicable and as listed in Table VII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 25 linear yards of nylon webbing. The lot shall be unacceptable if one (1) or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VII. Component and end item tests.

Characteristic	Requirement Reference	Test method
Warp, filling and binder yarn:		
Nylon identification	3.4.1	AATCC TM20 (see 6.3)
Denier	3.4.1.1	ASTM D1907
Ply		Visual
Twist	3.4.1.2	ASTM D1423
Visual shade matching	3.5.2	AATCC EP9 & 4.6.1
Spectral reflectance	3.6	4.6.2
Identification yarns:		
Fiber identification	3.7	AATCC TM20 (see 6.3)
Denier	3.7	ASTM D1907
Colorfastness of identification yarns:	Table I	
Light		AATCC TM16.3 1/
Laundering		AATCC TM61, Option 2, 2/
Colorfastness (webbing):	Table I	
Light		AATCC TM16.3, 1/
Laundering (3 cycles)		AATCC TM61, Option 2, 2/
Crocking		AATCC TM8 3/

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TABLE VII. Component and end item tests. - Continued

Characteristic	Requirement Reference	Test method
Weave (all Classes and Types)	3.8	Visual
Width, (inches)	Table IV and Table V	ASTM D3774, Option B
Thickness (inches)	Table IV and Table V	ASTM D1777
Weight, per linear yard	Table IV and Table V	ASTM D3776/D3776M, Opt C
Breaking strength Initial After abrasion (Types XXII and XXVII)	3.9.1, Table IV and Table V	ASTM D6775 ASTM D6770 & ASTM D6775
Ends in warp: Face and back Binder	Table IV and Table V	ASTM D3775
Picks per inch	Table IV and Table V	Visual
No. of single yarns for final plied yarn: Warp Binder Filling	Table IV and Table V	ASTM D3775
Curvature	3.9.2	4.6.3
Shrinkage	3.10	4.6.4
pH	3.11	AATCC TM81

1/ Rated using the AATCC EP1, Gray Scale for Color Change
2/ Rated using the AATCC EP1, Gray Scale for Color Change and AATCC EP2, Gray Scale for Staining.
3/ Rated using the AATCC EP8, AATCC 9-Step Chromatic Transference Scale

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching evaluation. The color and appearance of the dyed webbing shall match the applicable standard sample or color standard when viewed using the AATCC EP9, Option C with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (± 200) K.

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4.6.2 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometer (nm) at 20nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be viewed at an angle of no greater than 10º from normal, the specular component included. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage (excluding webbings). The specimen shall be measured as a single layer backed with layers of the same fabric and shade as follows: Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp and filling yarns or wales and courses when presented to sample port. The diameter for standard aperture size used in the color measurement device shall be 1/4 to 1/2 inches. For webbing greater than 1 1/4 inches in width, the diameter for standard aperture size used in the color measurement device shall be 1.0 to 1 1/4 inches in diameter. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.6.3 Measurement of lateral curvature.

4.6.3.1 Test specimen. The test specimen shall be a length of webbing, full width, measuring a minimum of 40 inches. The specimen shall not be stretched, smoothed or otherwise changed from its original condition prior to testing.

4.6.3.2 Number of determinations. Five (5) specimens shall be tested from each sample unit and averaged.

4.6.3.3 Apparatus.

Plexiglas (poly methyl methacrylic or PMMA) or equal – Pre-weighing approximately 35 ounces with dimensions of 45 inches by 5 inches by 1/4 inch
Straight edge - A rigid straight edge measuring 36 inches in length
Roller - A roller 1 inch in diameter and weighing 1½ pounds.

4.6.3.4 Procedure. The specimens shall be placed flat, on a smooth, horizontal flat surface without tension and allowed to reach moisture equilibrium as defined in the standard textile conditions as specified in ASTM D1776. After equilibrium is reached a weight shall be placed at one end of the webbing. The roller shall be placed on the specimen at the end of the webbing where the weight is located. The specimen should be approximately in the center of the roller. The roller shall be rolled along the length of the specimen, care being taken to keep the specimen in the center of the roller and not to exert any pressure on the roller. When the roller has passed the length of the webbing, the PMMA shall then be placed on the specimen for a period of one (1) hour. Without moving the PMMA on the specimen, the straight edge shall be placed on the

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Plexiglas so that both ends of the straight edge are aligned perpendicularly with the outermost edge of the specimen. Determine the highest degree of curvature of the specimen from the straight edge by measuring to the nearest 1/32 of an inch perpendicularly from the straight edge. Record the highest measurement (see Figure 9).

4.6.3.5 Report. The average of five (5) determinations from each sample unit shall be recorded.

4.6.4 Residual shrinkage test. Three 24-inch full width specimens shall be selected from the sample unit. The specimens shall be brought to equilibrium under standard conditions as defined in ASTM D1776. Eighteen-inch gage marks shall then be placed on the specimens with an indelible marking medium. The specimens shall then be placed in a container of water maintained at 100 (±2) °F and kept submerged at that temperature for 60 to 70 minutes.

Note: It may be necessary to attach weights to the specimens during the exposure period.) At the end of the exposure period, the specimens shall be removed and air dried without tension at a temperature not exceeding 105°F. After drying, the specimens shall be brought to equilibrium under standard conditions as defined in ASTM D1776, the distance between the gage marks re-measured and shrinkage calculated as follows:

$$\text{Percent of shrinkage} = \frac{(\text{Initial length} - \text{Final length})}{\text{Initial length}} \times 100$$

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The webbing is for use in the manufacturing of military tentage, clothing and equipage items used by the Department of Defense.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of webbing required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1)
e. Color of webbing required (see 3.5)
f. Conformance inspection acceptance quality limits (AQL) (see 4.3)
g. Inspection conditions (see 4.3.1)
h. Packaging (see 5.1)

6.3 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

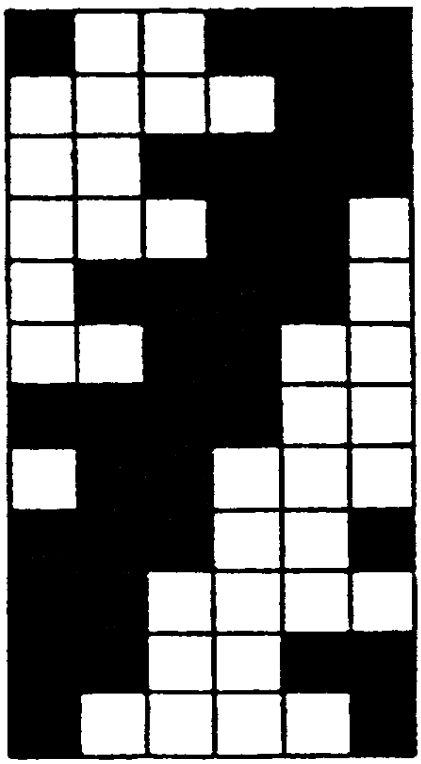
6.4 Color coding. A producer of these types of webbings, who is not listed in MIL-STD-1480, will apply to the U.S. Army Combat Capabilities Development Command Soldier Center, Natick, MA for assignment of color coding. When manufacturer's identification of other types is required, yarns colored as designated in MIL-STD-1480, or in the procurement documents, will be incorporated as specified by the U.S. Army Combat Capabilities Development Command Soldier Center, Natick, MA.

6.5 Subject term (key word) listing.

Binder
Catch-cord
Curvature

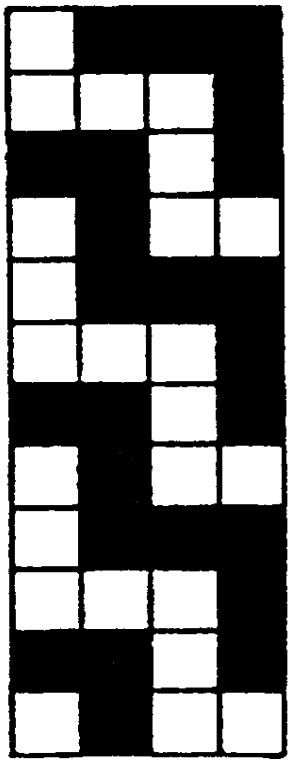
6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Body

One (1) repeat warp
One (1) repeat filling

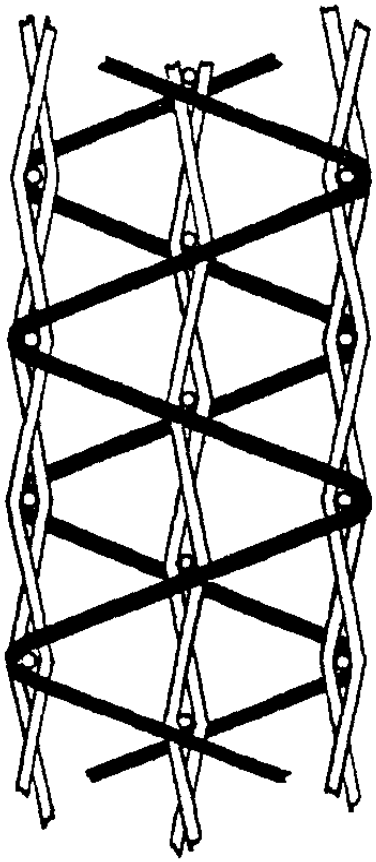


Selvage

One (1) repeat warp
Three (3) repeats filling

FIGURE 1. Weave Diagram for Types XIV, XVIII, XIX and XXI.

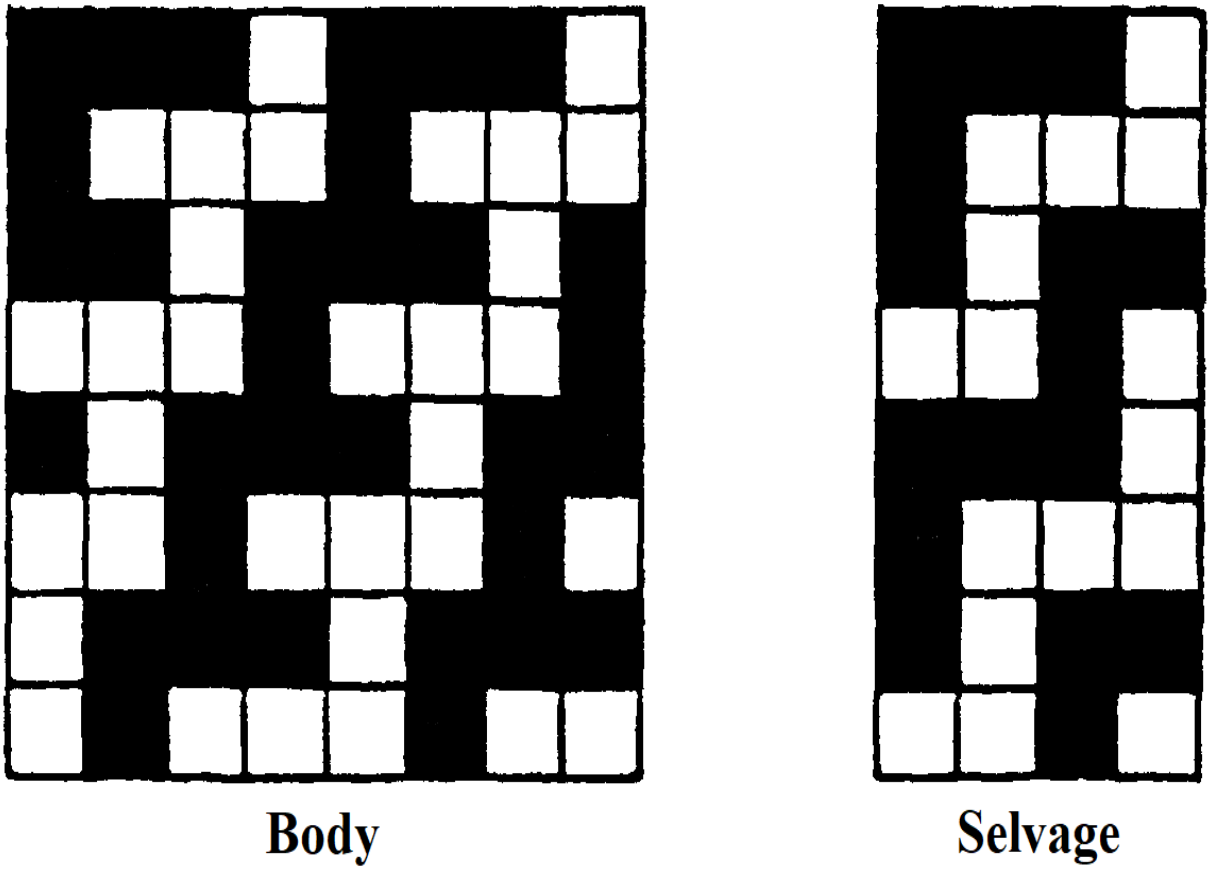
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Type XX - Warp yarns weaving one (1) end as one (1)
Type XXIII - Warp yarns weaving two (2) end as one (1)
Binder yarns both types weave one (1) end as one (1)

FIGURE 2. Cross section filling for Types XX and XXIII.

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Body Weave: **1/3 twill with the back filling. Two (2) ends weaving as one (1)**
Selvage Weave: **Double plain weave. Five (5) ends on one (1) edge; six (6) ends on other edge; one (1) end weaving as one (1)**

FIGURE 3. Weave diagram Type XXII.

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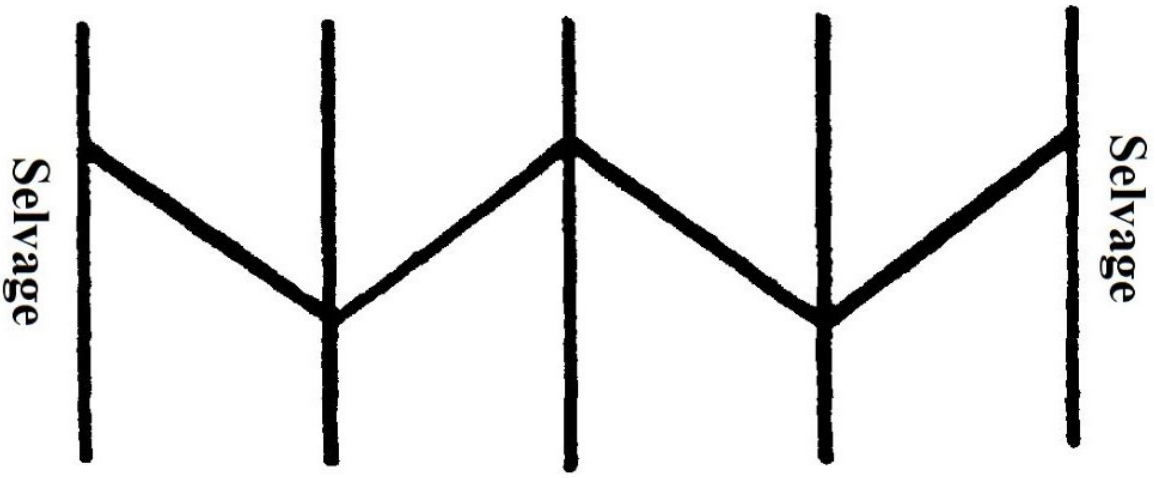


FIGURE 4. Weave diagram Type XXIV.

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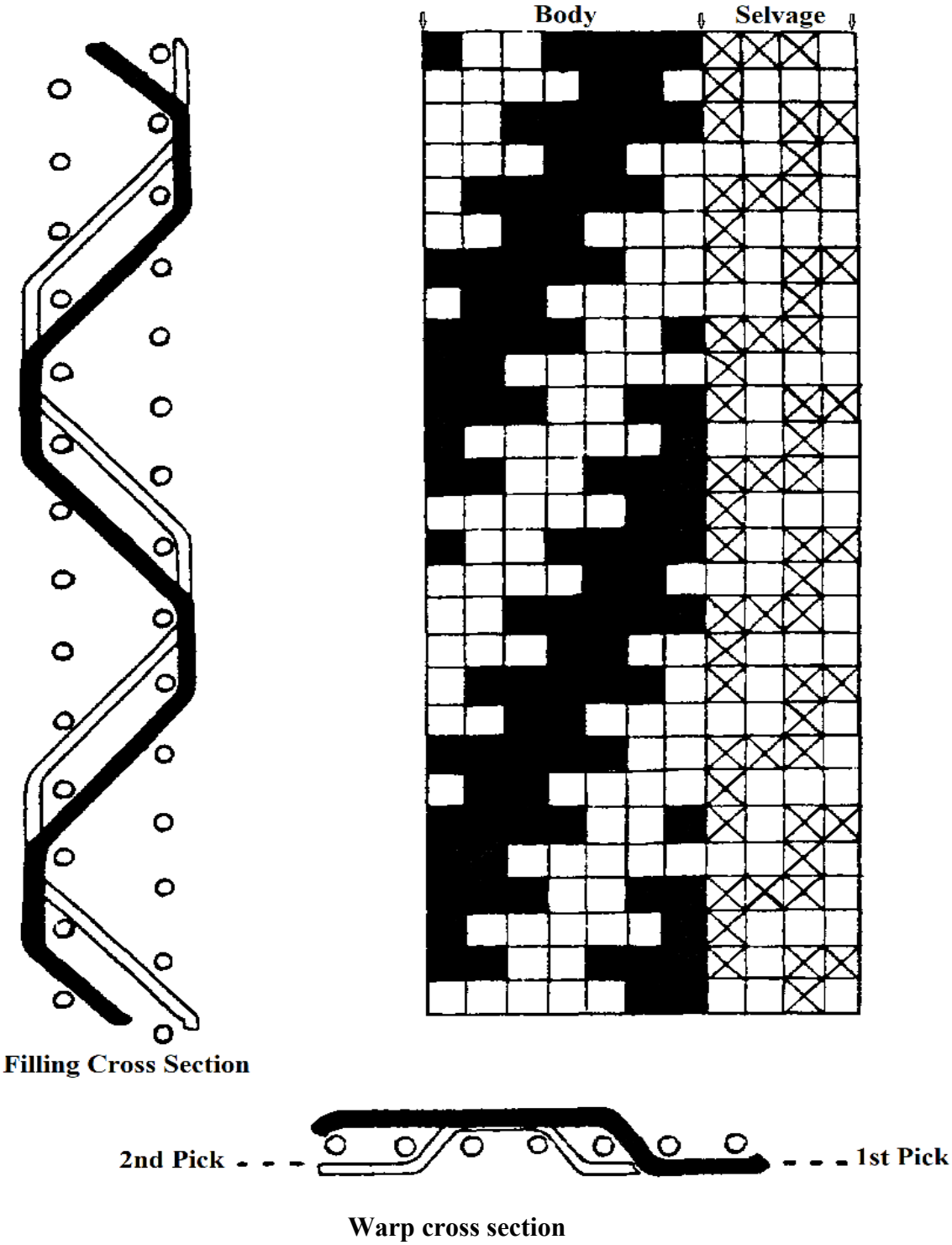


FIGURE 5. Weave diagram Type XXVI.

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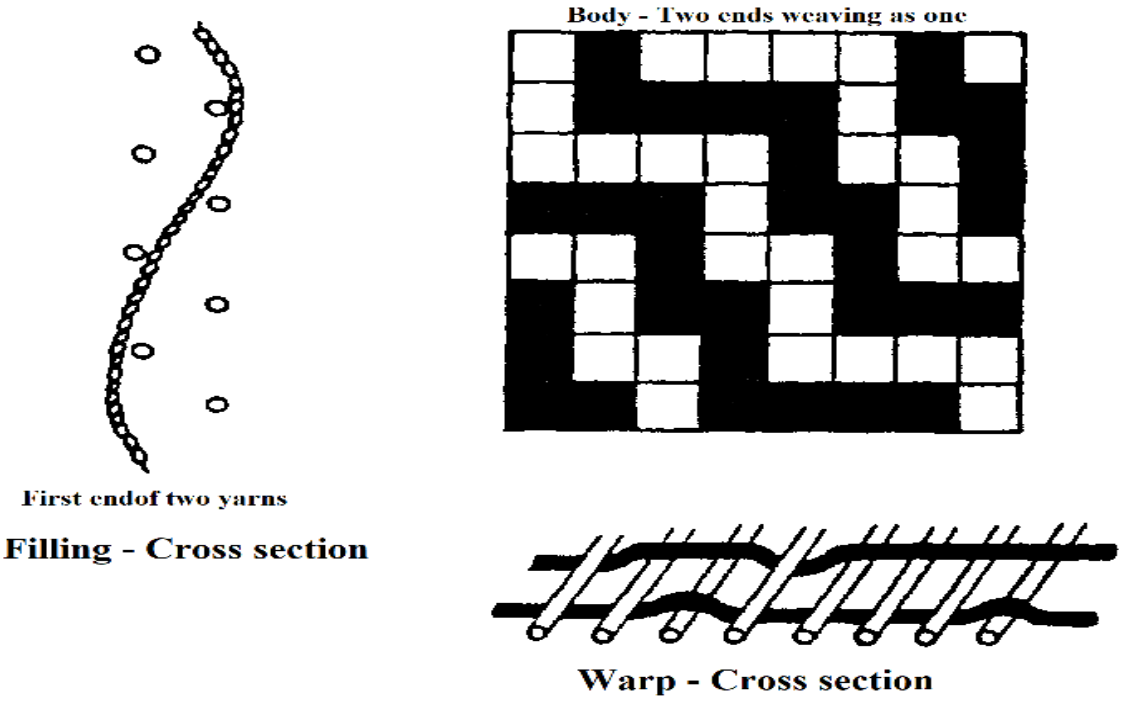


FIGURE 6. Weave pattern Type XXVII.

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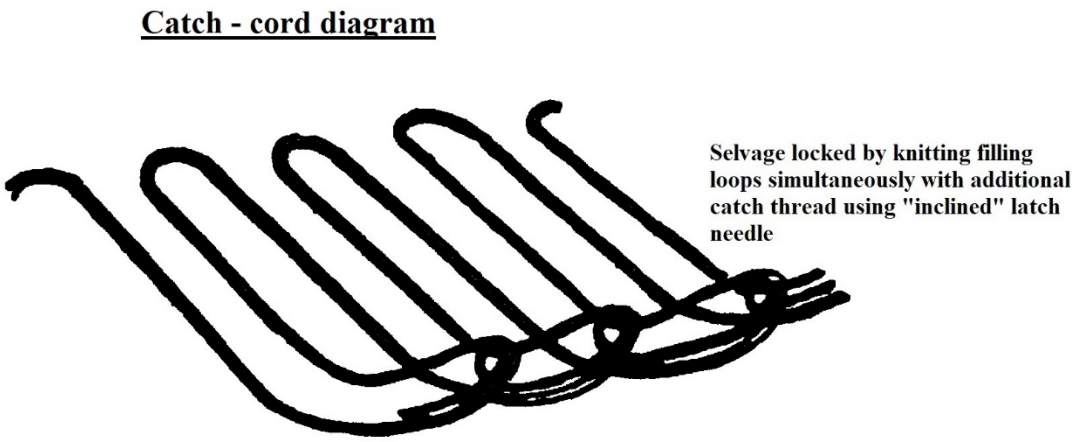


FIGURE 7. Catch-cord diagram for Classes 1A and 2.

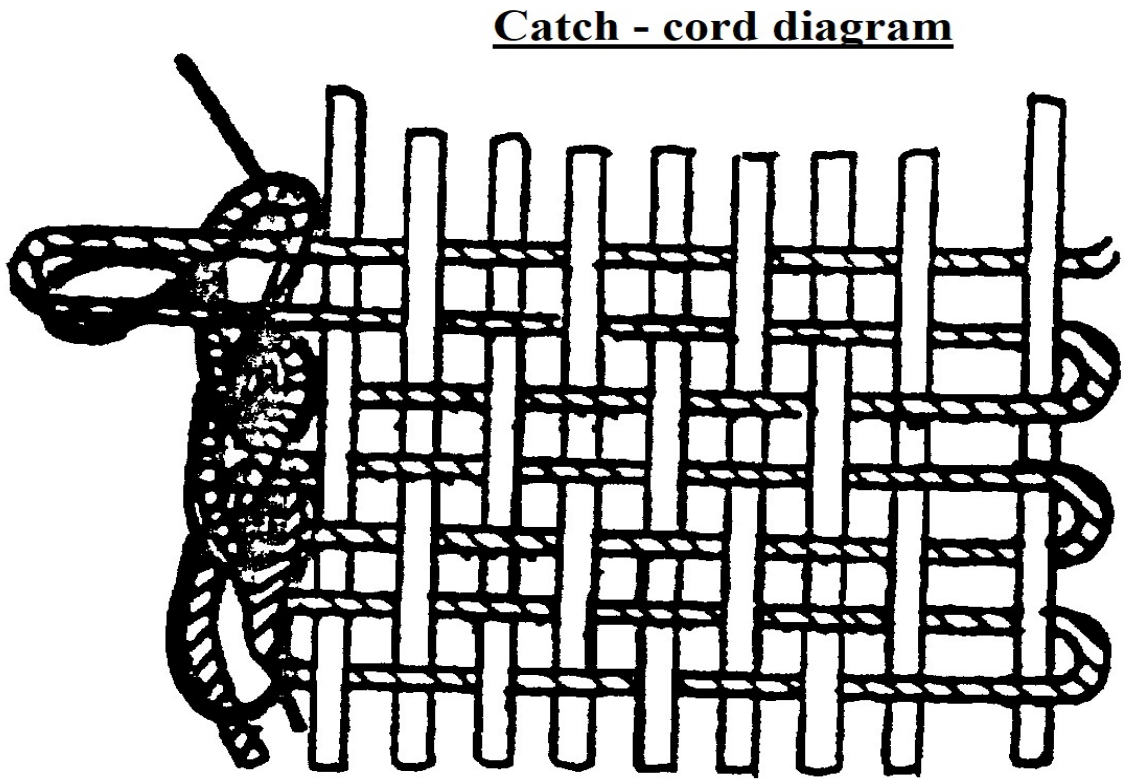


FIGURE 8. Another view of catch-cord diagram for Classes 1A and 2.

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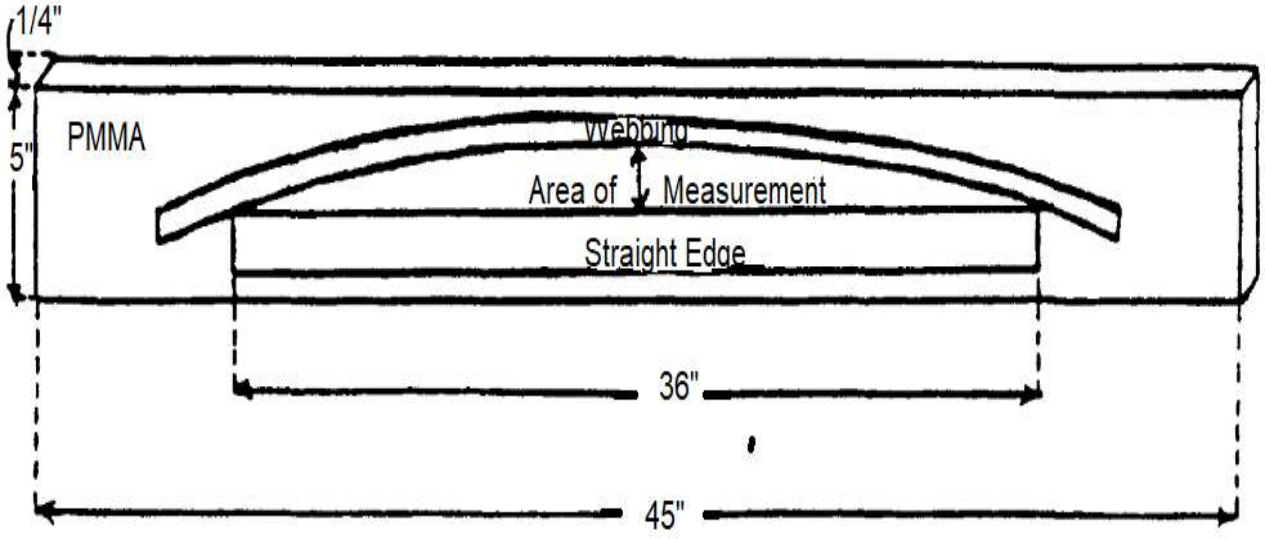


FIGURE 9. Diagram curvature measurement.

Custodians:
Army - GL
Navy - AS
Air Force - 11

Preparing activity:
DLA-CT

Review Activities:
Army – AT, MD, MI
Navy – MC, NU

(Project 8305-2021-007)

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